

Amir Ata Ghaffarian

CONTACT INFORMATION

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ABOUT ME

Computer Science graduate, passionate about Artificial Intelligence, Machine Learning. Experienced in building AI/ML systems and conducting research for real-world applications.

EDUCATION

Amirkabir University of Technology (Tehran Polytechnic)

Tehran, Iran

B.Sc in Computer Science

2020 – 2025

GPA: 4.0/4.0 (18.97/20) — Last Year GPA: 4.0/4.0 (19.60/20)

Relevant Coursework: Machine Learning, Deep Learning, Artificial Intelligence, Data Structures & Algorithms, Information Retrieval

HONORS

- Ranked in top **0.3%** (464/150,000) in National University Entrance Exam (Mathematics).
- Ranked in top **5%** (3rd) among Computer Science students at Amirkabir University.

RESEARCH INTERESTS

World Models & RL, Robotics, Embodied AI, LLMs, Neuroscience, Computational Biology

EXPERIENCES

Research Intern

Okinawa Institute of Science and Technology – Japan

Feb 2026 – Present

Currently a Research Intern at OIST's Genomics and Regulatory Systems Unit, working under the supervision of Prof. Nicholas Luscombe and Dr. Charles Plessy. My research lies at the intersection of AI and Bioinformatics, where I apply advanced Deep Learning architectures such as Self-Supervised Learning and Graph Neural Networks to decode genomic rearrangement patterns. By representing pairwise alignments through a multimodal approach, we aim to uncover structural variations at scale. I manage the end-to-end computational pipeline on an HPC cluster, utilizing Slurm and Singularity for seamless scalability and reproducibility.

DevOps engineer

ICT Group, Tehran, Iran

Aug 2025 – Jan 2026

Supported enterprise infrastructure by deploying and troubleshooting containerized services on Kubernetes and building monitoring and centralized-logging workflows with Prometheus, Grafana, Filebeat, and the Elastic Stack.

Site Reliability Engineering Intern

Neshan

Jul 2025 – Aug 2025

Completed an intensive bootcamp on Site Reliability Engineering, gaining hands-on experience in system administration, infrastructure automation, container orchestration, and monitoring systems.

Research Assistant

Center for Information Systems and Data Science

Jan 2025 – Jun 2025

Conducted research on predictive maintenance for batteries and bearings using transformers and diffusion models under the supervision of Dr. Babak Khalaj and Dr. Mohammad Hossein Rohban.

Software Engineering Intern

[Yektanet](#)

Jun 2024 – Jul 2024

Developed Dotanet Ad Server features (ad retrieval, auctions, click/impression tracking) with Go, Docker, PostgreSQL, and Kafka, ensuring scalability across multiple publisher websites.

Undergraduate Teaching Assistant

Amirkabir University & Sharif University

2022 – 2025

Machine Learning (Prof. Ali Sharifi-Zarchi, Sharif University),
Advanced Programming, AI, Algorithms (Prof. Mehdi Ghatte, Amirkabir University),
Image Processing (Prof. Mostafa Shamsi, Amirkabir University),
Operating Systems, Software Engineering (Prof. Zahra Ghorbanali, Amirkabir University)

PROJECTS

MyTorch – Deep Learning Framework [GitHub](#)

Built a from-scratch NumPy-based automatic-differentiation and neural-network framework with a PyTorch-like API, forward/reverse-mode AD, optimizers, training diagnostics, and reproducible run tracking. Used agentic coding partner and validated generated changes through 1,200+ Python tests, gradient checks, strict type/lint gates, multi-version CI, benchmarks, and detailed English technical documentation.

Knowledge Graph Generator [GitHub](#) : Developed a system to generate knowledge graphs from various data sources using Python. Integrated with FalkorDB for graph storage, Kafka for real-time data processing, MinIO for object storage, Redis for ensuring idempotency.

DiffRUL-CMAPSS [GitHub](#)

Built a diffusion-based data augmentation model (PyTorch) for aero-engine RUL estimation, integrating LSTM, transformers, and DDPM, based on a 2024 RESS paper.

Search Engine [GitHub](#)

Designed an IR system (Python, TF-IDF, ParsBERT) with K-Means clustering and FAISS for efficient document retrieval.

Dotanet [GitHub](#)

Created a scalable ad platform (Go, PostgreSQL, Kafka) with RESTful APIs for ad serving, auctions, and real-time tracking.

SKILLS

Programming Languages Python, Golang, C/C++, Java

Frameworks & Libraries NumPy, Pandas, TensorFlow, PyTorch, Scikit-learn, OpenCV
, NLTK, FastAPI, Gin, Django

Databases MySQL, PostgreSQL, Redis, FalkorDB, MinIO

Tools Linux, Git, Docker, Docker Compose, Kubernetes, Slurm, Singularity,
Kafka, Prometheus, Grafana, Nginx, Gitlab CI/CD, ELK Stack

CERTIFICATES & WORKSHOPS

[Machine Learning Specialization](#)

DeepLearning.AI, Stanford University, Issued May 2024

[Neuromatch Deep Learning Course](#)

Neuromatch, Issued Jul 2025

[Foundational Cloud Practitioner](#)

Cloud Academy, Issued Sep 2025

[Project-Oriented Course In Backend Development With Golang](#)

Quera, Issued Dec 2024

[Task-Oriented Course in Linux](#)

Quera, Issued Mar 2024

[Advanced Algorithmic Thinking and Data Structure](#)

Quera, Issued Oct 2023